Natalyx MVP Fixes

Work Package 07: Commercial configuration, enrollment, and promotion

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1 Work Package 07: Commercial configuration, enrollment, and promotion

Review draft only. Ownership and sequencing remain recommendations until Luke and Alan approve the plan together.

Plan	Value
Work package	WP07
Plan file	`WP07 - Commercial configuration, enrollment, and promotion.md`
Proposed PRs	P07-01, P07-02, P07-03, P07-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1.1 Purpose and outcome

Problem. The business model requires versioned price/terms, explicit commitment, and one precise processor-neutral event before workspace-to-case promotion; none is currently authoritative.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned commercial configuration and acceptances, a separate commitment/enrollment lifecycle, one processor-neutral byos.workspace_promotion_authorized.v1 event, and exactly-once paid-case creation by its consumer.

Explicit non-goals.

- No processor-specific webhook
- No automated unresolved pricing policy
- No activation before onboarding evidence
- No browser-redirect authority

1.2 Verified current-state evidence

Classification is explicit: **verified** is observed code/PR/live behavior; **verified limitation** records an evidence gap; **design authority** is an approved planning source; **design decision** is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

1.3 Architecture and contracts

- Every commitment, invoice, payment, credit, refund, and promotion records the activated commercial-configuration version.
- Only the app-owned commitment/commercial domain emits byos.workspace_promotion_authorized.v1 after durable funding/terms/credit validation; a processor callback or redirect never emits it.
- The consumer creates/fetches one enrollment_pending case and restricted IP membership; GC case access follows the paid-case invitation contract.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`case`, `case_invitation`, `journey_commitment`, `node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Read active config and record acceptance commands. Create, accept, cancel, inspect enrollment. Internal event producer only. Internal event handler only.
Storage	Commercial configuration and acceptance ledgers. Commitment events/current projection. Promotion ledger and source-workspace unique constraint.

Contract area	Package rule
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

1.3.1 Applicable state machines

1.3.2 Node 0 workspace (`node0_workspace`)

Owner: **natalyx-app**. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

Transition: `none` -> `draft`

- **Actor:** authenticated IP
- **Evidence:** first explicit Node 0 start
- **Idempotency key:** `workspace:{owner}:create`
- **Invalid transitions:** Existing active workspace returns existing
- **Audit event:** `workspace_created`
- **Retry:** Unique owner/idempotency key
- **Recovery:** Forward repair orphan membership

Transition: `draft` -> `active`

- **Actor:** IP
- **Evidence:** minimum workspace consent/profile
- **Idempotency key:** `workspace:{id}:activate:{revision}`
- **Invalid transitions:** Missing consent/owner
- **Audit event:** `workspace_activated`
- **Retry:** Return current active revision
- **Recovery:** Correct inputs and retry

Transition: `active` -> `match_ready`

- **Actor:** IP + selected GC evidence policy
- **Evidence:** selected candidate + required provisional/GC-authored fields
- **Idempotency key:** `workspace:{id}:ready:{revision}`

- **Invalid transitions:** Critical unknown/decline/revoked claim
- **Audit event:** `workspace_match_ready`
- **Retry:** No duplicate event
- **Recovery:** Reopen active with reason revision

Transition: `match_ready -> promotion_pending`

- **Actor:** promotion event consumer
- **Evidence:** valid byos.workspace_promotion_authorized.v1
- **Idempotency key:** `promotion:{commitment}:v1`
- **Invalid transitions:** No authorization; already promoted; stale config
- **Audit event:** `workspace_promotion_requested`
- **Retry:** Replay resumes/finds same promotion
- **Recovery:** Forward-repair incomplete transaction

Transition: `promotion_pending -> promoted`

- **Actor:** promotion service
- **Evidence:** paid case + manifest committed
- **Idempotency key:** `promotion:{commitment}:v1`
- **Invalid transitions:** Projection invariant/privacy failure
- **Audit event:** `workspace_promoted`
- **Retry:** Return same case ID
- **Recovery:** Repair from immutable manifest; never duplicate

Transition: `draft|active|match_ready -> archived`

- **Actor:** IP/operator per policy
- **Evidence:** explicit archive + retention policy
- **Idempotency key:** `workspace:{id}:archive:{request}`
- **Invalid transitions:** Promotion pending/promoted
- **Audit event:** `workspace_archived`
- **Retry:** Idempotent
- **Recovery:** Restore only through new audited revision if policy permits

1.3.3 Commercial commitment (`journey_commitment`)

Owner: **natalyx-app commercial domain**. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

Transition: `none -> draft`

- **Actor:** IP
- **Evidence:** active workspace + displayed config/terms
- **Idempotency key:** `commitment:{workspace}:{config}`
- **Invalid transitions:** No active config/workspace
- **Audit event:** `commitment_drafted`

- **Retry:** Return existing draft
- **Recovery:** Create new draft after expiry/version change

Transition: `draft -> accepted`

- **Actor:** IP
- **Evidence:** terms/privacy consent + payment path
- **Idempotency key:** `commitment:{id}:accept:{terms_hash}`
- **Invalid transitions:** Stale/missing acceptance
- **Audit event:** `commitment_accepted`
- **Retry:** Return accepted
- **Recovery:** New acceptance revision if version changes

Transition: `accepted -> funding_pending`

- **Actor:** commercial service
- **Evidence:** enrollment invoice/payment intent created
- **Idempotency key:** `commitment:{id}:funding`
- **Invalid transitions:** No invoice or invalid amount
- **Audit event:** `funding_requested`
- **Retry:** Resume same invoice
- **Recovery:** Cancel/reissue under audited adjustment

Transition: `funding_pending -> funded`

- **Actor:** payment domain
- **Evidence:** net settled enrollment funding meets versioned requirement
- **Idempotency key:** `enrollment-funding:{id}:v1`
- **Invalid transitions:** Redirect/pending/failed/reversed evidence
- **Audit event:** `commercial.enrollment_funding_confirmed.v1`
- **Retry:** Exactly-once event/outbox replay
- **Recovery:** Reconcile inbox/ledger; no browser repair

Transition: `draft|accepted|funding_pending -> cancelled`

- **Actor:** IP/authorized operator
- **Evidence:** explicit cancellation and refund policy
- **Idempotency key:** `commitment:{id}:cancel:{request}`
- **Invalid transitions:** Already funded/promoted uses case cancellation rail
- **Audit event:** `commitment_cancelled`
- **Retry:** Idempotent
- **Recovery:** New commitment required

Transition: `draft|accepted -> expired`

- **Actor:** system

- **Evidence:** offer/terms TTL or config deactivation
- **Idempotency key:** `commitment:{id}:expire`
- **Invalid transitions:** No expiry after funding
- **Audit event:** `commitment_expired`
- **Retry:** Idempotent
- **Recovery:** Draft new version

1.3.4 Paid case lifecycle (`case`)

Owner: **natalyx-app**. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

Transition: `none` -> `enrollment_pending`

- **Actor:** promotion service
- **Evidence:** byos.workspace_promotion_authorized.v1 + projection manifest
- **Idempotency key:** `promotion:{commitment}:v1`
- **Invalid transitions:** No authorization/duplicate workspace
- **Audit event:** `paid_case_created`
- **Retry:** Return same case
- **Recovery:** Forward-repair same promotion

Transition: `enrollment_pending` -> `active`

- **Actor:** authorized coordinator/domain service
- **Evidence:** approved onboarding/compatibility/intake/education evidence
- **Idempotency key:** `case:{id}:activate:{evidence_hash}`
- **Invalid transitions:** Missing/revoked evidence or GC consent
- **Audit event:** `case_activated`
- **Retry:** No duplicate activation/invoice
- **Recovery:** Return pending with unmet checklist

Transition: `active` -> `paused`

- **Actor:** authorized commercial/operations policy
- **Evidence:** audited non-critical pause reason
- **Idempotency key:** `case:{id}:pause:{event}`
- **Invalid transitions:** Safety-critical access cannot be paused
- **Audit event:** `case_paused`
- **Retry:** Idempotent
- **Recovery:** Resume after evidence; preserve entitled docs

Transition: `paused` -> `active`

- **Actor:** authorized operator/domain service
- **Evidence:** pause reason resolved
- **Idempotency key:** `case:{id}:resume:{event}`
- **Invalid transitions:** Unresolved safety/authorization block

- **Audit event:** `case_resumed`
- **Retry:** Idempotent
- **Recovery:** Escalate manual review

Transition: `active|paused -> closed`

- **Actor:** authorized coordinator
- **Evidence:** birth/close or completed termination handoff evidence
- **Idempotency key:** `case:{id}:close:{evidence_hash}`
- **Invalid transitions:** Open critical obligations
- **Audit event:** `case_closed`
- **Retry:** No-op if same evidence
- **Recovery:** Reopen only by exceptional audited event

Transition: `enrollment_pending|active|paused -> cancelled`

- **Actor:** authorized operator under policy
- **Evidence:** withdrawal/non-clearance/cancellation + financial disposition
- **Idempotency key:** `case:{id}:cancel:{request}`
- **Invalid transitions:** Cannot erase records/refund ledger
- **Audit event:** `case_cancelled`
- **Retry:** Idempotent
- **Recovery:** Replacement is a new case/workspace policy event

1.3.5 Paid-case membership invitation (`case_invitation`)

Owner: **natalyx-app**. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

Transition: `none -> issued`

- **Actor:** authorized IP case member
- **Evidence:** enrollment_pending/active case + recipient binding
- **Idempotency key:** `case-invite:{case}:{role}:{generation}`
- **Invalid transitions:** Self/member/active-GC/unauthorized
- **Audit event:** `case_invite_issued`
- **Retry:** Return current; supersede prior live GC invite
- **Recovery:** Issue corrected invite

Transition: `issued -> pending_identity`

- **Actor:** recipient
- **Evidence:** valid addressed token landing
- **Idempotency key:** `case-invite:{id}:land`
- **Invalid transitions:** Expired/revoked
- **Audit event:** `case_invite_opened`
- **Retry:** Safe repeat

- **Recovery:** Authenticate with return_to

Transition: `pending_identity -> identity_bound|mismatch_pending`

- **Actor:** auth provider
- **Evidence:** verified subject/email
- **Idempotency key:** `case-invite:{id}:bind:{subject}`
- **Invalid transitions:** Terminal/duplicate/different subject
- **Audit event:** `case_invite_bound|mismatch`
- **Retry:** CAS protects terminal
- **Recovery:** Issuer confirmation or corrected invite

Transition: `mismatch_pending -> issuer_confirmed`

- **Actor:** authorized issuer
- **Evidence:** explicit reviewed mismatch confirmation
- **Idempotency key:** `case-invite:{id}:confirm:{issuer}`
- **Invalid transitions:** Issuer unauthorized/recipient changed
- **Audit event:** `case_invite_mismatch_confirmed`
- **Retry:** CAS no-op
- **Recovery:** Revoke and reissue

Transition: `identity_bound|issuer_confirmed -> consented`

- **Actor:** invitee
- **Evidence:** paid-case role/privacy consent version
- **Idempotency key:** `case-invite:{id}:consent:{version}`
- **Invalid transitions:** Stale/missing consent
- **Audit event:** `case_invite_consented`
- **Retry:** Idempotent
- **Recovery:** Present current consent

Transition: `consented -> redeemed`

- **Actor:** invitation service
- **Evidence:** single transaction creates person/party/membership
- **Idempotency key:** `case-invite:{id}:redeem`
- **Invalid transitions:** Existing member/active GC/expired/revoked
- **Audit event:** `case_invite_redeemed`
- **Retry:** Return same membership
- **Recovery:** Rollback entire transaction; forward repair from invite

Transition: `issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented -> declined|revoked|expired|superseded`

- **Actor:** invitee/issuer/system

- **Evidence:** explicit decline/revoke/TTL/new generation
- **Idempotency key:** `case-invite:{id}:terminal:{event}`
- **Invalid transitions:** No terminal overwrite
- **Audit event:** `case_invite_terminal`
- **Retry:** Idempotent CAS
- **Recovery:** Fresh invitation only

1.3.6 Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

1.3.7 Sole promotion-authorizing event

`event_type: byos.workspace_promotion_authorized.v1`

Producer: natalyx-app commitment/commercial domain after deterministic validation. Consumer: natalyx-app promotion service.

Payload:

```
event_id
idempotency_key
workspace_id
commitment_id
owner_person_id
owner_subject
commercial_config_version
terms_acceptance_ids[]
payment_plan_id
funding_evidence.kind
funding_evidence.commercial_event_id
funding_evidence.settled_amount_minor
funding_evidence.currency
credit_allocation_ids[]
initial_case_state=enrollment_pending
occurred_at
```

`correlation_id`
`causation_id`
`schema_version`

Authorization conditions:

- Workspace is active/match-ready, unarchived and not already promoted.
- Commitment references the same workspace, has current terms/privacy acceptances and is in funding_pending/funded per the approved state contract.
- Payment path and activated commercial configuration are pinned.
- Required enrollment funding has a verified settled ledger entry or separately approved financing/manual-funding evidence; a processor webhook is first normalized and cannot itself authorize promotion.
- Credit allocation is non-negative, reproducible and cannot exceed the applicable obligation.
- No cancellation, compliance hold, reversal that invalidates funding, or existing promotion is present.

Unique constraints:

- one promotion per workspace
- one case per promotion event
- one result per idempotency key
- one primary owner membership per created case

No browser success/return route and no raw processor callback can emit or substitute for this event.

1.4 Explicit PR sequence

Order	ID	Title	Repository	Dependencies	Gate	Size	Readiness
1	P07-01	Add versioned commercial configuration and terms acceptance	app	P02-03	A2	Large	Dependencies unmet
2	P07-02	Implement commitment and enrollment-pending lifecycle	app	P07-01, P08-01	A2	Large	Dependencies unmet
3	P07-03	Emit processor-neutral workspace promotion authorization	app	P04-02, P07-02, P08-02	A2	Large	Dependencies unmet
4	P07-04	Promote workspace exactly once from authorization event	app	P03-03, P04-02, P07-03	A2	Exceptional	Dependencies unmet

1.4.1 P07-01 - Add versioned commercial configuration and terms acceptance

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P02-03
Blocking decisions/checkpoints	D-COM-01, D-COM-02, D-COM-03
Source / salvage	Payment/business-model DOCX

1.4.1.1 Objective Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.

1.4.1.2 Scope

- Immutable activated config versions
- JMF/deposit/discount/milestones/credit policy
- Terms and privacy acceptance records

1.4.1.3 Explicit non-goals

- No unresolved Node 0 price value
- No payment provider

1.4.1.4 Expected files or modules

- `migrations/versions/*_commercial_configuration.py`
- `app/domain/commercial_config.py`
- `app/storage/commercial_config.py`
- `tests/python/test_commercial_config.py`

1.4.1.5 Change map

Area	Planned change
Schema/model	Commercial configuration and acceptance ledgers.
API/routes	Read active config and record acceptance commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.1.6 Tests and validation

Objective tests:

- Activation immutability
- currency/amount validation
- acceptance version
- no retroactive mutation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.1.7 Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.1.8 Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.1.9 Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.1.10 Dependencies and parallel work

Hard dependencies: P02-03. Parallel opportunities: P07-03, P07-04

1.4.1.11 Objective acceptance criteria

- ☐ Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.1.12 Ownership and review

Role	Recommendation
DRI	Luke

Role	Recommendation
Implementer	Luke (proposed; pending joint approval)
Reviewer	Alan
Final approver	Joint
Required expertise	commercial systems, PostgreSQL
Higher-capability review	Mandatory before merge

1.4.2 P07-02 - Implement commitment and enrollment-pending lifecycle

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P07-01, P08-01
Blocking decisions/checkpoints	D-COM-04
Source / salvage	Business-model conversion trigger

1.4.2.1 Objective Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.

1.4.2.2 Scope

- Draft/accepted/cancelled/expired commitment transitions
- Payment-path choice
- Enrollment invoice command

1.4.2.3 Explicit non-goals

- No browser-triggered promotion
- No activation

1.4.2.4 Expected files or modules

- `app/domain/commitments.py`
- `app/storage/commitments.py`
- `app/routes/enrollment.py`
- `tests/python/test_commitments.py`

1.4.2.5 Change map

Area	Planned change
Schema/model	Commitment events/current projection.
API/routes	Create, accept, cancel, inspect enrollment.

Area	Planned change
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.2.6 Tests and validation

Objective tests:

- Acceptance prerequisites
- duplicate accept
- stale config
- cancel/expire
- resume

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.2.7 Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.2.8 Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.2.9 Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.2.10 Dependencies and parallel work

Hard dependencies: P07-01, P08-01. Parallel opportunities: P07-04

1.4.2.11 Objective acceptance criteria

- ☐ Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.
- ☐ Every listed objective test passes locally and at the required gate where applicable.

- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.2.12 Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending joint approval)
Reviewer	Alan
Final approver	Joint
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

1.4.3 P07-03 - Emit processor-neutral workspace promotion authorization

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P04-02, P07-02, P08-02
Blocking decisions/checkpoints	D-COM-04, D-DATA-01
Source / salvage	Business-model promotion invariants

1.4.3.1 Objective Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.

1.4.3.2 Scope

- Deterministic authorization predicate
- Canonical payload and idempotency key
- Transactional event/outbox
- Funding/credit/config evidence references

1.4.3.3 Explicit non-goals

- No case creation
- No redirect/webhook-as-authority
- No GC membership

1.4.3.4 Expected files or modules

- `app/domain/promotion_authorization.py`

- `app/event_handlers/commercial.py`
- `app/storage/outbox.py`
- `tests/python/test_promotion_authorization.py`

1.4.3.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal event producer only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only commercial-domain authority can emit the typed event.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.3.6 Tests and validation Objective tests:

- Pending/failed/reversed funding
- stale terms/config
- duplicate/replayed inputs
- credit mismatch
- event payload/checksum

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.3.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.3.8 Rollout Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.3.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.3.10 Dependencies and parallel work Hard dependencies: P04-02, P07-02, P08-02. Parallel opportunities: P07-01

1.4.3.11 Objective acceptance criteria

- ☐ Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.3.12 Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending joint approval)
Reviewer	Alan
Final approver	Joint
Required expertise	event-driven systems, payments, commercial policy
Higher-capability review	Mandatory before merge

1.4.4 P07-04 - Promote workspace exactly once from authorization event

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P03-03, P04-02, P07-03
Blocking decisions/checkpoints	D-COM-04, D-DATA-01, CHECK-ALAN-122-124
Source / salvage	byos.workspace_promotion_authorized.v1 contract, PR #124 source linkage concepts

1.4.4.1 Objective Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.

1.4.4.2 Scope

- Unique promotion idempotency key
- Case/source link and config version
- Purpose-limited data manifest and audit event

1.4.4.3 Explicit non-goals

- No redirect authority

- No GC case membership transfer
- No activation

1.4.4.4 Expected files or modules

- `app/domain/promotion.py`
- `app/storage/promotions.py`
- `app/event_handlers/enrollment.py`
- `tests/python/test_workspace_promotion.py`

1.4.4.5 Change map

Area	Planned change
Schema/model	Promotion ledger and source-workspace unique constraint.
API/routes	Internal event handler only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Projection defaults deny and preserves provenance.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.4.6 Tests and validation Objective tests:

- Replay/concurrency
- partial transaction
- funding event validation
- private candidate exclusion
- GC invitation required

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.4.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.4.8 Rollout Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.4.9 Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.4.10 Dependencies and parallel work

Hard dependencies: P03-03, P04-02, P07-03. Parallel opportunities: P07-01, P07-02

1.4.4.11 Objective acceptance criteria

- ☐ Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.4.12 Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending joint approval)
Reviewer	Alan
Final approver	Joint
Required expertise	event-driven systems, privacy, PostgreSQL
Higher-capability review	Mandatory before merge

1.5 Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P07-01` → `P07-02` → `P07-03` → `P07-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

1.6 Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Terms/config version pinning; Commitment retries; Exact-once promotion; Funding reversal handling; No GC access escalation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

1.7 Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Treatment
Blocking/product/compliance decision	D-DATA-01 - Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?	Luke	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	D-DATA-01 - Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Blocking/product/compliance decision	D-COM-01 - Set the flat one-time price and tax treatment; it must remain fully creditable against the later JMF.	Luke	P06-01, P07-01, P08-01; Before activation of a Complete configuration	D-COM-01 - Keep no active production value until explicitly approved; test with synthetic configuration.
Blocking/product/compliance decision	D-COM-02 - Ratify $8,900 JMF, $7,830 pay-in-full, $750 deposit and 20/30/20/20/10 post-deposit milestones as configuration v1 with effective date/currency.	Luke	P07-01, P08-01, P08-03; Before P07-01 activation data	D-COM-02 - Use the document anchors but require explicit version activation rather than code literals.
Blocking/product/compliance decision	D-COM-03 - Approve the actual paid-service terms, privacy/consent disclosure and refundable-to-earned wording.	Luke	P07-01, P07-02, P08-03; Before P07-02 user acceptance	D-COM-03 - External legal/privacy/finance review before any real acceptance.

Type	ID / risk	Owner	Affected PRs / latest point	Treatment
Blocking/product/compliance decision	D-COM-04 - Define evidence for enrollment funding confirmed, promotion authorization, and for onboarding/compatibility complete enough to move case active and earn/issue activation billing.	Luke	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	D-COM-04 - Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.
Blocking/product/compliance decision	D-COM-05 - Select financing partner and define when funding is sufficiently authorized/settled; otherwise hide the path.	Luke	P07-02, P08-02; Before financed option is exposed	D-COM-05 - Do not block card/ACH MVP; keep optional and disabled.
Implementation risk	Configuration drift	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Duplicate promotion	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Redirect spoofing	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Unresolved policy encoded as default	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

1.8 Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	-	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Payment/business-model DOCX as authority
- PR #124 source workspace linkage concepts only

1.9 Ownership recommendation

Luke leads this work package, with Alan providing technical review. The split is proposed for review and can be amended before implementation begins.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P07-01	Luke	Luke (proposed)	Alan	Joint	Mandatory
P07-02	Luke	Luke (proposed)	Alan	Joint	Mandatory
P07-03	Luke	Luke (proposed)	Alan	Joint	Mandatory
P07-04	Luke	Luke (proposed)	Alan	Joint	Mandatory

Luke leads implementation and product acceptance for this package; Alan reviews architecture, migration, security and failure behavior. This is a recommendation pending joint review, not a compulsory assignment.

1.10 Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and review record complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.